

Investigating Users' Inclination of Leveraging Mobile Crowdsourcing to Obtain Verifying vs. Supplemental Information when Facing Inconsistent Smart-city Sensor Information

You-Hsuan Chiang
National Yang Ming Chiao Tung
University
Department of Computer Science
Hsinchu, Taiwan
youxuanjiang.cs07@nycu.edu.tw

Tzu-Yu Huang
National Yang Ming Chiao Tung
University
Department of Computer Science
Hsinchu, Taiwan
hajime.cs09@nycu.edu.tw

Je-Wei Hsu
National Yang Ming Chiao Tung
University
Department of Computer Science
Hsinchu, Taiwan
jewehsu.cs10@nycu.edu.tw

Hsin-Lun Chiu
National Central University
Department of Information
Management
Taoyuan, Taiwan
107403009@cc.ncu.edu.tw

Chung-En Liu
National Yang Ming Chiao Tung
University
Department of Computer Science
Hsinchu, Taiwan
n09142002.cs09@nycu.edu.tw

Yung-Ju Chang
National Yang Ming Chiao Tung
University
Department of Computer Science
Hsinchu, Taiwan
armuro@cs.nycu.edu.tw

ABSTRACT

Smart cities leverage sensor technology to monitor urban spaces in real-time. Still, discrepancies in sensor data can lead to uncertainty about city conditions. Mobile crowdsourcing, where on-site individuals offer real-time details, is a potential solution. Yet it is unclear whether users would prefer to utilizing the mobile crowd on site to verify sensor data or to provide supplementary explanations for inconsistent sensor data. We conducted an online experiment involving 100 participants to explore this question. Our results revealed a negative correlation between perceived plausibility of sensor information and inclination to use mobile crowdsourcing for obtaining information. However, only around 80% of participants relied on crowdsourcing when they felt the sensor information as implausible. Interestingly, even when participants believed the sensor data to be plausible, they sought to use crowdsourcing for further details half of the time. We also found that participants leaned more towards using the crowd for explanations (48% and 49% of instances) rather than seeking verification when encountering perceived implausible sensor information (38% and 32% of instances).

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; **Empirical studies in collaborative and social computing**.

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KEYWORDS

smart city, information seeking, mobile crowdsourcing, sense-making, information consistency, plausibility, sensor plausibility

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1 INTRODUCTION

The validity and reliability of data in smart cities are crucial to ensuring accurate and useful information for citizens and travelers [15, 17]. Nonetheless, sensor-generated data can be flawed - incomplete, inaccurate, inconsistent, or unclear [1, 20], due to variances in the sensed objects or sensor characteristics [18] and abstraction methods [3]. Mobile crowdsourcing, with smartphone-equipped users gathering diverse location-based information, presents a possible solution. Workers can assess and interpret local conditions, enhancing sensor data with high-quality, contextually rich information (e.g., [2, 5, 7, 10, 12, 21]). This makes mobile crowdsourcing a potential tool for individuals to alleviate uncertainty arising from inconsistent sensor readings. But, would people take advantage of a mobile crowd to lessen this uncertainty, if such a service is available? And how frequently? Would they use it for verification [9] or explanation of contradictory sensor data [4, 14], which may be more prone to errors and time-consuming but can aid users in better grasping the on-site conditions? Our study aims to tackle these questions. Specifically, we designed two types of crowdsourcing information providing tasks, *verification* and *supplementary explanation*, which require different levels of effort and deliver varying degrees of informational detail. The verification task focuses on

confirming or refuting existing data, while the supplementary explanation task offers additional insights and contextual information beyond what verification provides. For instance, in the context of real-time bus information, the displayed data on remaining arrival time and bus location may not be consistent. Verification information relies solely on passengers' perspectives, providing their current location and estimated arrival time. In contrast, supplementary explanation goes beyond verification data and offers further explanations for bus delays, such as traffic congestion or accidents. Our research questions are:

RQ1: *Would individuals choose to use a mobile crowdsourcing service for on-site information when confronted with sensors providing inconsistent data?*

Furthermore, to examine the preference for the type of information people are inclined to obtain from the crowd, we inquire:

RQ2: *Would users prefer the on-site crowd to confirm the sensor data or supply supplementary explanations?*

2 METHODOLOGY

To address our research questions, we employ a theoretical framework called the Plausibility Gap Model (PGM) proposed by Klein et al.[13] to manipulate the crowdsourcing situations. The framework posits that perceiving lower plausibility in the presented information about a situation leads to larger gaps, consequently increasing an individual's uncertainty towards the situation. Such uncertainty would prompt people to seek information to resolve the perceived uncertainty [8, 16]. In this study, we assume that a future smart city is equipped with diverse sensors that provide multiple real-time data streams, which may display different information that could lead individuals to perceive the information as *inconsistent*. Previous research has established that information consistency plays a crucial role in shaping plausibility judgments[6, 11, 19]. Therefore, we manipulate plausibility by altering the consistency between two sensor information sources, both of which can be inferred by the individual to understand the environment. Consequently, as depicted in Figure 1, we first hypothesize that:

- H1: The perceived consistency between the information from two different sensors is positively correlated with the perceived plausibility of the situation.

Subsequently, assuming that lower plausibility would make individuals more likely to seek additional information to help them resolve the uncertainty, we hypothesize that:

- H2a: The likelihood of intending to use mobile crowdsourcing to obtain external information is negatively correlated with the perceived plausibility of the situation.

Additionally, we assume that individuals would perceive the amount of information detail between verification and supplementary information differently. When individuals perceive the situation as less plausible, they would be inclined to acquire more information (e.g., seeking supplementary explanation rather than verification) to help them make sense of the situation. Thus, we hypothesize that:

- H2b: When intending to use crowdsourcing, the likelihood of obtaining additional detailed information is associated with the plausibility of the situation.

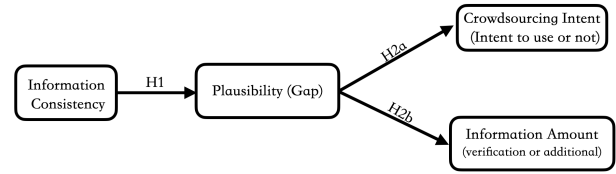


Figure 1: Model Diagram

2.1 Experiment Design

This study employed a scenario-based online approach combined with the "think-aloud" method[22], followed by an interview to collect both quantitative and qualitative data. A remote online format was utilized to ensure standardization and convenience for participants.

2.1.1 Scenario Design. The study presented participants with a set of 20 scenarios related to smart city applications, including people, parking, and traffic. These scenarios were categorized into "crowd (cars) and seat (parking spot) occupancy" and "real-time waiting time and location," with five scenarios each for locations such as the library, gym, restaurant, table tennis hall, parking space, intercity bus, short-distance bus, train, ship, and high-speed rail.

Each scenario, as shown in Figure 2a, provided two related pieces of information from two separate sensors, manipulated for information consistency. To prevent participants from interpreting the scenarios differently due to their varied prior experiences of similar situations, which might significantly affect their perception of plausibility and inclination to use additional information, we incorporated information about their existing expectations toward the environment (e.g., expecting the space to be crowded) and the urgency of the situation into the scenarios to control for these situational factors. Doing so not only allowed us to reduce the impact of their interpretation due to prior experience, but also enabled us to observe how these two factors influenced their choices. In this paper, due to limited length, we focus on testing the aforementioned hypotheses and do not investigate the impact of these situational factors.

Below the description of each scenario was a list of seven questions, as depicted in Figure 2b, with the first three questions serving as a manipulation check for urgency, information consistency, and pre-existing expectations. Participants then assessed the plausibility of two presented sensor, followed by questions evaluating their inclination to obtain external information from people on-site via a mobile crowdsourcing service, and any changes they would make when considering different factors, such as monetary cost and potential waiting time. However, we focused solely on scenarios that did not involve practical factors that could potentially alter participants' decisions. Therefore, the data from questions 6 and 7 were not considered in the present study.

2.1.2 Experiment Procedure. During the study, participants were presented with a web page containing 20 scenarios in a questionnaire format. They were instructed to engage in "think-aloud" practices to allow us to gain insights into their cognitive processes, decision-making strategies, and the rationales behind them. The

Scenario 1

Recently moved to a new city and decided to go to a nearby chain restaurant for a meal this afternoon.

1. If you want to dine in, you must arrive at the restaurant within half an hour at the latest, or you may not be able to finish your meal before the restaurant closes.

2. Since you have no idea how crowded it will be at this time, you first check the real-time information of the ramen restaurant.

3. **Crowd in the Restaurant**
Instant Number of People : 40 People
The Restaurant can Accommodate 45 People

Seat Occupancy

Occupied Non-occupied

(a)

1. How urgently do you want to know the on-site situation based on the scenario above?
Not Urgent ○○○○○○ Extremely Urgent

2. How consistent do you think the two types of real-time information are?
Extremely Inconsistent ○○○○○○ Extremely Consistent

3. Which of the following options is closest to your thoughts?
☐ Both real-time information are similar to what I expected
☐ Only real-time people count is similar to what I expected
☐ Only seat usage status is similar to what I expected
☐ Both real-time information are different from what I expected
☐ I did not have any expectations

4. Which of the following options best describes the on-site situation according to you?
☐ I believe the on-site situation is similar to both types of real-time information
☐ I believe the on-site situation is only similar to real-time people count information
☐ I believe the on-site situation is only similar to seat usage status
☐ I believe the on-site situation is different from both types of real-time information
☐ I am not sure

5. Assuming that you can now use the app to ask on-site people to answer questions without considering point costs and waiting time, what kind of information do you want to get?
☐ I want to get verified information
☐ I want to get additional supplementary information
☐ I do not need any other information

6. Considering the waiting time required to obtain information and the urgency of the situation, what information do you want on-site people to provide?
☐ I want on-site people to provide me with verified information
☐ I want on-site people to provide me with additional supplementary information
☐ I do not need any assistance from others

7. Considering point costs, since it may take more effort to supplement than to verify, it may require more basic points to attract people to answer. How would you spend your points to get information?
☐ Provide what I consider to be the basic verification points and wait for verification
☐ Spend more verification points to reduce the waiting time for verification
☐ Provide what I consider to be the basic supplementary points and wait for supplementation
☐ Spend more supplementary points to reduce the waiting time for supplementation
☐ I am not willing to spend points

(b)

Figure 2: An example of the online study page presenting a scenario along with the seven standard questions. The (a) scenario section contains manipulations pertaining to (1) urgency, (2) pre-existing expectation, and (3) information consistency. The (b) question section comprises items that evaluate (4) manipulation checking, (5) plausibility measurement, and (6) the participants' inclination to seek external information under different levels of practical consideration.

study was conducted online via a conference call; therefore, participants were instructed to share their screens and verbalize their thoughts while answering the questions.

Upon completion of the scenarios, we conducted a brief debriefing interview with participants to clarify or ask about their responses during the study that we found as interesting patterns or relatively different from other participants, in order to investigate any additional factors they considered. Questions included why they would be inclined to obtain additional information in certain situations, why they preferred verification vs. supplementary explanation in different situations, and so on. This approach allowed for the identification of any ambiguous or unclear responses and provided a deeper understanding of the reasoning behind participants' choices. The entire study for each participant took approximately one and a half to two hours per participant to complete.

2.2 Participants Recruitment and Data Collection

We recruited 100 participants aged 20-60 in Taiwan via social media platforms, comprising 40 males and 60 females, aged between 20 and 59. Of these, 70 participants fell in the age group of 20-29, 18 in the age group of 30-39, 10 in the age group of 40-49, and 2 in the age group of 50-59. Each participant receives a reward of NT\$400 (approximately 13 USD). The experiment was recorded and transcribed for further analysis, using rigorous and dependable methods.

2.3 Data Cleaning and Analyze

We implemented manipulation checks to ensure the validity of the data collected. For analysis, we utilized a mixed-effect logistic regression model in which participant ID was included as a random effect to examine the associations between information consistency, plausibility, and the inclination and amount of external information obtained through mobile crowdsourcing.

3 RESULT

3.1 Sensor Consistency vs. Plausibility

The results show that participants' perception of the plausibility of the sensors was positively correlated with the consistency between the sensors. As depicted in Figure 3a, where plausibility was rated in three levels (2: both sensors were perceived plausible; 1: one sensor was perceived plausible; 0: no sensor was perceived plausible), participants' perceptions of plausibility increased as the consistency of the information increased. The regression analysis also shows a positive correlation between consistency and plausibility ($Z=19.62, p<.001$). Thus, H1 is supported. In particular, 76% of the time, participants perceived that information from both sensors was plausible when the two sensors' information was perceived consistent, whereas only 2% of the participants perceived that information from both sensors was plausible when the two sensors' information was perceived inconsistent. However, it is noteworthy that, in the latter situation, participants slightly tended to lean toward thinking that both were implausible over thinking either one is plausible (51% vs. 47%), though the difference was not significant. Notably, 22% of the time, participants thought both sensor information was implausible even when there was consistency between the two sensors. This finding suggests that even when facing consistent information, participants still were likely to suspect or question the correctness of the information. Further qualitative analysis is needed to obtain more information about this outcome.

3.2 Did Perception of Low Plausibility lead to the Inclination of Using Crowdsourcing?

To address our second research question, we examined the effect of perceived plausibility on individuals' decisions to utilize crowdsourcing for obtaining external information. Results indicate that individuals are more inclined to obtain external information when they perceive the real-time sensor information to be implausible. As shown in Figure 3b, the proportion of participants not inclined to use crowdsourcing to obtain external information increased with

higher plausibility (14%, 19%, and 48%). Furthermore, the proportion of individuals who opted for the "explanation" option when using crowdsourcing was low when they perceived the sensor information to be highly plausible (25% when they perceived the information from both sensors to be plausible). The regression results revealed a negative correlation between perceived plausibility and both the inclination to obtain external information and the likelihood of choosing "explanation" as their action (inclination to obtain external information: $Z=-15$, $p<.001$; likelihood of choosing explanation: $Z=-4.107$, $p<.001$). Thus, H2a and H2b are supported.

However, interestingly, the likelihood of participants wanting to obtain supplemental explanations (i.e., getting more information) from on-site individuals was nearly equal (49% vs. 48%) between situations where participants deemed that both sensors were implausible and those where they deemed that only one of the two sensors was plausible. We suspect that this might be because participants wanted to get an explanation whenever they distrusted one of the sensors, perhaps attempting to make sense of why the information was inconsistent. Notably, even when participants perceived the information as highly plausible, with trust in both sensors' information, more than half of them (52%) demonstrated an inclination to seek external information through crowdsourcing. This unexpected finding indicates that the motivation to utilize mobile crowdsourcing for additional information is not solely influenced by perceived plausibility. Further qualitative analysis is necessary to gain insights into this intriguing phenomenon.

4 DISCUSSION

Based on the results, we propose two findings that could potentially influence crowdsourcing in the context of smart cities: 1) Despite high sensor consistency, external information remains valuable to users. This could imply that individuals anticipate corroborating their understanding of on-site situations via diverse sources of information, rather than relying on a single type of information source. 2) When sensor consistency is low, the demand for supplemental explanations through crowdsourcing increases. Participants facing inconsistent information may be more likely to seek reasons for these disparities, leading to a higher propensity for crowdsourced supplemental information. However, qualitative analysis is still required to understand the reasoning behind user choices in different scenarios.

5 CONCLUSION AND FUTURE WORK

In this study, we investigated individuals' inclination to leverage mobile crowdsourcing to obtain information about an environment when they faced sensor information that sensed that environment displaying inconsistent information in a smart city context. We have demonstrated the positive correlation between consistency of information between sensors and plausibility, as well as between plausibility and inclination to leverage mobile crowdsourcing to get information. These findings suggest that mobile crowdsourcing is a promising approach for providing additional information for individuals to obtain environmental information in addition to sensors. However, further analysis is required to explore the influence of situational factors, including pre-existing expectations

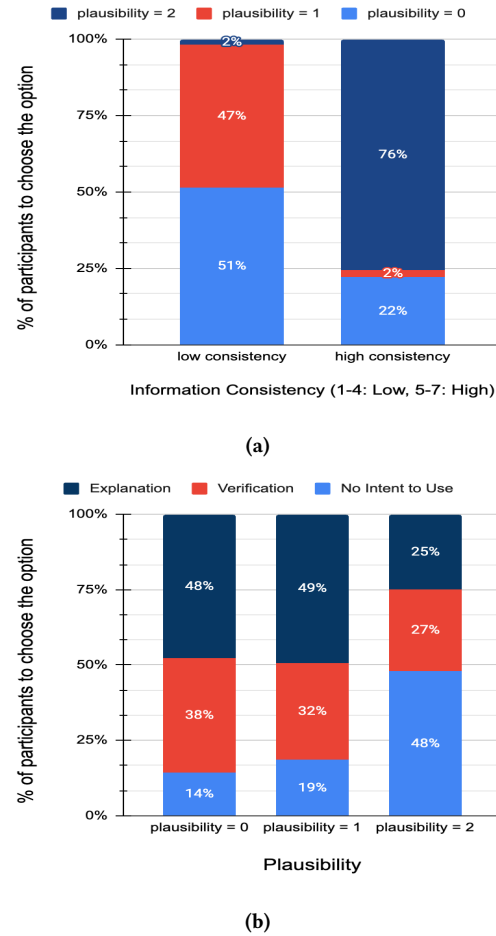


Figure 3: Proportion of Participants' Selection, (a) Consistency to Plausibility, (b) Plausibility to Crowdsourcing Choice

and perceived urgency, on the perceived plausibility of sensor information. Additionally, practical considerations such as the cost and waiting time associated with crowdsourcing need to be examined to understand individuals' inclination to utilize mobile crowdsourcing. Qualitative analysis is also necessary to gain insights into the motivations underlying participants' crowdsourcing choices and the factors influencing these motivations.

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